

Andres Ramirez-Jaime

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Professional Summary

AI/ML Research Scientist at the Chan Zuckerberg Biohub (Inverse Problems & Machine Learning), developing machine learning and generative methods for inverse problems in cryo-electron microscopy and tomography (cryoEM/cryoET). Ph.D. in Electrical and Computer Engineering (University of Delaware) focused on multimodal, physics-informed generative AI for computational imaging and perception—with an emphasis on diffusion models and transformers for compressive satellite LiDAR using SAR and imaging spectroscopy side information, and extensions to hyperspectral/SAR remote sensing and gigapixel pathology. Co-authored several peer-reviewed publications and contributed to funded research with NASA and NSF partners, with a research trajectory focused on building reliable, high-impact AI systems that connect modern generative modeling with sensing and decision pipelines. Industry research experience includes Apple (LLM tool-use/agenic workflows: RAG, LoRA fine-tuning, large-scale evaluation and test-suite design) and Vertex Pharmaceuticals (semantic segmentation for whole-slide pathology), demonstrating the ability to translate AI advances into cross-domain impact. Extensive teaching experience as an instructor across programming, robotics, control, machine learning, and statistics, paired with mentoring and outreach.

Education

University of Delaware

Ph.D. in Electrical and Computer Engineering

Newark, DE, USA

Jun 2026

- GPA: 3.952

- Dissertation: Computational Lidar From Space: Volumetric Representations, Generative Reconstruction, and Joint Source-Channel Coding.

University of Los Andes

M.S. in Computer and Electronic Engineering

Bogota, Colombia

Mar 2016

- Graduated *Cum Laude*.

University of Los Andes

B.S. in Electronic Engineering

Bogota, Colombia

Oct 2013

Research Experience

Chan Zuckerberg Biohub

AI/ML Research Scientist, Inverse Problems & Machine Learning

Redwood City, CA, USA

Jul 2026 – Present

- Research on machine learning and generative methods for inverse problems in cryo-electron microscopy and tomography (cryoEM/cryoET).

University of Delaware (NASA CASALS Project)

Research and Teaching Assistant

Newark, DE, USA

Feb 2022 – Jun 2026

- Lead research on generative and multimodal ML for LiDAR and hyperspectral remote sensing (diffusion models, transformers, GANs) within the NASA CASALS project, contributing to advancements in AI and computer vision.
- Developed reconstruction and super-resolution methods for the HyperHeight Data Cube (HHDC) 2m dataset (~100k 3D tensors) using a novel forward imaging model; achieved <1m MAE on CHM/DTM and multiple height percentiles under compressive sampling (~25% sampling), demonstrating strong physics-informed ML capabilities.
- Built scalable end-to-end pipelines (data processing → training → evaluation) on ~60 GB datasets; ran ~5-day training cycles on a single A100 and produced inference over ~600 m² areas in ~12 min, showcasing efficient deep learning model deployment.
- Published and presented research: **24 peer-reviewed papers** (9 first-author); presented at international venues in remote sensing and computational imaging, highlighting strong communication skills.
- Mentored 2 Ph.D. students, 1 master's student, and visiting researchers; fostering collaboration skills.

University of Delaware

Visiting Scholar

Newark, DE, USA

Jul 2021 – Dec 2021

- Designed the HyperHeight Data Cube (HHDC) representation for efficient storage and processing of compressed 3D satellite LiDAR data, enabling downstream learning and reconstruction workflows.

- Implemented a 3D convolutional autoencoder for HHDC reconstruction, improving reconstruction quality vs. classical image-processing baselines by **+6 dB** on CHM and **+18 dB** on DTM.

Robotics and Industrial Informatics Institute (UPC)

Visiting Researcher

Barcelona, Spain

Sep 2015 – Mar 2016

- Investigated real-time control techniques based on game theory for urban drainage systems; applied methods to Barcelona's sewer network, achieving results comparable to model predictive control approaches.

University of Los Andes

CIFI Researcher

Bogotá, Colombia

Feb 2015 – Feb 2016

- Conducted research on urban drainage and climate change under the "Urban Drainage and Climate Change: Towards the Sewerage Systems of the Future - PAVCO Phase II" project; implemented real-time control techniques for flood mitigation in urban areas.

Teaching Experience

University of Delaware

Teaching Assistant

Newark, DE, USA

Feb 2022 – Jun 2026

- Taught undergraduate courses including Imaging and Deep Learning, and Probability/Statistics, and graduate courses including Analytics I: Statistical Learning.

University of La Sabana

Mechanical Engineering Professor

Chia, Colombia

Jan 2018 – Dec 2021

- Taught undergraduate courses including Introduction to Programming, Digital Logic, Robotics I, Electrotechnics, Dynamic Systems and Control, Electronic Instrumentation, Electric Machines, and Introduction to Machine Learning.
- Supervised 12 undergraduate thesis projects; taught courses in prototypes and manufacturing.

National Learning Service (SENA)

Robotics Facilitator

Soacha, Colombia

Feb 2017 – Dec 2017

- Taught robotics and electronic hardware to apprentices from vulnerable populations at the TecnoAcademy Cazucá node; co-led the Material, Manufacturing Process, and Automation Research Group (MAPRA), mentoring students on applied robotics projects.

Industry Experience

Apple Inc.

Large Language Models and Generative AI Engineering Intern

Cupertino, CA, USA

Jun 2024 – Aug 2024

- Built a retrieval-augmented generation (RAG) system grounded in the iPhone user manual to generate higher-quality internal tool descriptions; improved descriptions for **~85%** of tools by strengthening retrieval and grounding.
- Fine-tuned Apple Intelligence LLMs for iPhone-focused tool usage via dataset curation and LoRA adapters on a distributed GPU cluster (up to **96 GPUs**); improved single-turn tool-selection accuracy from **61%** to **90.2%**.
- Created evaluation datasets and hand-crafted test suites (**~2,500** human-created conversations spanning **~80** tools) to stress-test edge cases; used results to diagnose failure modes and prioritize fixes.
- Optimized the training/evaluation pipeline to increase iteration rate from **1** to **>4** cycles/day, accelerating experimentation and readiness for deployment.
- Reduced tool hallucinations (nonexistent tools) from **~3%** to **0%** on internal testing by analyzing error patterns and recommending mitigation changes.

Vertex Pharmaceuticals

Computer Vision and Machine Learning Engineer

San Diego, CA, USA

Jun 2023 – Aug 2023

- Partnered with biologists, physicians, and chemists on early drug discovery (IPF, ADPKD), translating scientific goals into ML deliverables and evaluation criteria, demonstrating strong collaboration skills and domain knowledge in healthcare.
- Built a U-Net semantic segmentation model for gigapixel pathology images achieving **>90%** accuracy; replaced outsourced processing (100 slides; up to 3-month turnaround) with in-house inference (**~15 min** per processed slice/region), reducing vendor dependence and avoiding up to **\$600k/year** external spend.
- Developed an automated pipeline to segment kidney organoids in whole-slide images and estimate morphology (size, wall thickness) with **<3%** error, improving measurement consistency for downstream analysis.

Invited Talks

- “High-Altitude Earth Observation with Diffusion Models for Satellite LiDAR Reconstruction.” Electronic Imaging 2025, Burlingame, CA, USA, February 2–6, 2025.

Awards and Honors

- 2024 Doctoral Fellowship for Excellence, University of Delaware.
- 2024 ECE Signal Processing Award, University of Delaware.
- 2022 George W. Laird Fellow, University of Delaware.
- Graduated *Cum Laude*, M.S., University of Los Andes, 2016.

Open-Source Data & Software

- **HHDC Dataset (Hugging Face Hub)**: Created/maintained HyperHeight Data Cube dataset for LiDAR denoising/super-resolution; includes train/val/test splits and standardized 3D tensor format. huggingface.co/datasets/anfera236/HHDC
- **MatSWMM Toolbox**: Co-authored open-source toolbox for real-time urban drainage control simulations. github.com/gandresr/MatSWMM

Grants and Funding

- Co-authored successful proposal for NSF EAGER IMPRESS-U Grant #2404740: “Exploratory Research on Generative Compression for Compressive Lidar” (PI: Gonzalo R. Arce, University of Delaware; joint with ONRG and partners in Ukraine and Poland; \$300,000; 2024–2026).
- Co-authored successful proposal for UNIDEL210-UD AI CTR-ARCE PILOT: “Computational Lidar Sensing for Earth Science” (PI: Gonzalo R. Arce, Co-PI: R. Vargas, University of Delaware; \$15,000; 2/1/2023 – 1/30/2024).
- Co-authored successful proposal for NASA Grant #80NSSC22K1099: “Sampling and Deep Learning Reconstruction of LiDAR Waveforms” (PI: Gonzalo R. Arce, University of Delaware; \$40,000; 6/1/2022 – 5/31/2023).
- Co-authored successful proposal for NASA EPSCoR RID Grant: “Planetary LiDAR Compressive Point-Cloud Sampling and Machine Learning High-Resolution Reconstruction” (PI: Gonzalo R. Arce, Co-PI: Amo Tong, University of Delaware; \$28,000; 8/1/2021 – 6/30/2022).

Professional Service

- Reviewer, IEEE International Geoscience and Remote Sensing Symposium (IGARSS), 2024–2026.
- Peer Reviewer for The Journal of Supercomputing, IET Image Processing, and Journal of Spatial Science.

Leadership and Outreach

- Designed and programmed embedded control for the UNISABANA HERONS mechanical ventilator deployed in **400+** units during COVID-19; implemented respiratory control and data collection in C and LabVIEW, supporting regulatory approval by the Colombian FDA (University of La Sabana, 2018–2021).
- Led a robotics team for RoboCup 2019 (Sydney); reached the Community Shield final as runner-up, owning end-to-end technical execution (perception, control, systems integration) as team leader (University of La Sabana, 2018–2021).
- Active member, Hispanic/Latino Graduate Student Association (HLGSA), University of Delaware.
- Active member, Society for Advancement of Chicanos/Hispanics & Native Americans in Science (SACNAS).

Technical Skills

- **Programming:** Python, C, MATLAB, LabVIEW.
- **ML / Deep Learning:** PyTorch, TensorFlow, Keras; diffusion models, transformers, GANs; LLM fine-tuning (LoRA); retrieval-augmented generation (RAG); multimodal neural networks.
- **Computer Vision / Imaging:** semantic segmentation, super-resolution, denoising, reconstruction; computational imaging; medical and remote-sensing imagery.
- **Data / Systems:** large-scale training and evaluation (gigapixel images; 3D tensors; GPU training); dataset curation; experiment evaluation and test-suite design.
- **Tools:** Linux, Git, GIS, LaTeX.
- **Languages:** English (fluent), Spanish (native).

Publications

- Full list and citations available on Google Scholar.

Journal Articles

- Diffusion-Based Joint Recovery, Denoising, and Super-Resolution of Compressed-Sensing Satellite LiDAR Data. Ramirez-Jaime, A., Porras-Diaz, N., Stephen, M., Yang, G., Arce, G. R., Accepted, IEEE Transactions on Computational Imaging, 2026.
- Objective and Subjective Quality Assessment of Forest Landscapes Reconstructed Using Generative Diffusion Models From Compressed Satellite LiDAR Data. Ieremeiev, O., Lukin, V., Ramirez-Jaime, A., Arce, G. R., Kopytek, M., Lech, P., Okarma, K. P., IEEE Access, 2025.
- Toward Submeter Satellite Surface Topography and Vegetation Mapping Using LiDAR/RGB Constrained Generative Diffusion. Porras-Diaz, N., Ramirez-Jaime, A., Arce, G. R., Stephen, M., IEEE Transactions on Geoscience and Remote Sensing, 2025.
- On the Lossless Compression of HyperHeight LiDAR Forested Landscape Data. Makarichev, V., Ramirez-Jaime, A., Porras-Diaz, N., Vasilyeva, I., Lukin, V., Arce, G., Okarma, K., Remote Sensing, 2025.
- Generative diffusion models for compressed sensing of satellite lidar data: Evaluating image quality metrics in forest landscape reconstruction. Ramirez-Jaime, A., Arce, G. R., Porras-Diaz, N., Ieremeiev, O., Rubel, A., Lukin, V., Kopytek, M., Lech, P., Fastowicz, J., Okarma, K., Remote Sensing, 2025.
- Super-Resolved 3D Satellite Lidar Imaging of Earth Via Generative Diffusion Models. Ramirez-Jaime, A., Porras-Diaz, N., Arce, G. R., Stephen, M., IEEE Transactions on Geoscience and Remote Sensing, 2025.
- Transformer end-to-end optimization of compressive LiDARs using imaging spectroscopy side information. Porras-Diaz, N., Ramirez-Jaime, A., Arce, G. R., Pena-Pena, K., Harding, D., Stephen, M., MacKinnon, J., Vargas, R., IEEE Transactions on Geoscience and Remote Sensing, 2024.
- The development and implementation of a low-cost mechanical ventilator in a low-middle-income country during the COVID-19 pandemic: The Unisabana-HERONS. Giraldo-Cadavid, L. F., Echeverry, J., Varón-Vega, F., Bastidas, A., Ramirez-Jaime, A., Cardona, A. F., Vega, C. J. L., Serrano-Mayorca, C. G., Garay, D., Rincón, D. N., and others, Heliyon, 2024.
- HyperHeight LiDAR Compressive Sampling and Machine Learning Reconstruction of Forested Landscapes. Ramirez-Jaime, A., Pena-Pena, K., Arce, G. R., Harding, D., Stephen, M., MacKinnon, J., IEEE Transactions on Geoscience and Remote Sensing, 2024.
- Compressive spectral imaging via misalignment induced equivalent grayscale coded aperture. Zhang, T., Zhao, S., Ma, X., Ramirez-Jaime, A., Zhao, Q., Arce, G. R., IEEE Geoscience and Remote Sensing Letters, 2023.
- Positive-pressure ventilation: Modeling, validation, and stochastic control of mean-field type. Ramirez-Jaime, A., Barreiro-Gomez, J., Longas, L., Echeverry-Mejia, J., Tembine, H., Automatica, 2022.
- MatSWMM—an open-source toolbox for designing real-time control of urban drainage systems. Riaño-Briceño, G., Barreiro-Gomez, J., Ramirez-Jaime, A., Quijano, N., Ocampo-Martinez, C., Environmental Modelling & Software, 2016.

Book Chapters

- Computational Lidar from Space: Toward the Next Generation of Satellite Lidars. Ramirez-Jaime, A., Arce, G. R., Porras-Diaz, N., Stephen, M., Yang, G. In Handbook of Statistics, Vol. 54: Multidimensional Signal Processing (A. S. R. Srinivasa Rao, K. V. Mishra, and G. R. Arce, Eds.). Elsevier, 2026 (In Press).

Conference Proceedings

- Lossless Compression of Hyperheight LiDAR Forested Landscapes Data. Makarichev, V., Lukin, V., Vasilyeva, I., Okarma, K., Ramirez-Jaime, A., Arce, G. R., in Proc. 29th International Conference on Methods and Models in Automation and Robotics (MMAR), 2025.
- Denoising and Super-Resolution of Satellite Lidars Using Diffusion Generative Models. Ramirez-Jaime, A., Porras-Diaz, N., Arce, G. R., Stephen, M., in Proc. IEEE Statistical Signal Processing Workshop (SSP), 2025.
- Optimal illumination pattern for satellite compressive LiDAR using generative AI. Newman-Sanders, C., Ramirez-Jaime, A., Porras-Diaz, N., Arce, G. R., Stephen, M., in Proc. Machine Learning from Challenging Data 2025, 2025.
- SpectralCam: High-Resolution Low-Cost Spectral Imaging Using DSLR Cameras. Paruchuri, A., Ramirez-Jaime, A., Arce, G., Alrushud, A., Ma, X., Radpour, R., in Proc. IEEE ICASSP, 2025.
- Super-resolution of satellite lidars for forest studies via generative adversarial networks. Ramirez-Jaime, A., Porras-Diaz, N., Arce, G. R., Harding, D., Stephen, M., MacKinnon, J., in Proc. IEEE IGARSS, 2024.
- Multi-modal transformer for compressive LiDARs using hyperspectral imaging side-information. Porras-Diaz, N., Ramirez-Jaime, A., Arce, G. R., Vargas, R., Harding, D., Stephen, M., MacKinnon, J., in Proc. IEEE IGARSS, 2024.
- Super-resolution of satellite lidars for forest studies using diffusion generative models. Ramirez-Jaime, A., Arce, G. R., Stephen, M., MacKinnon, J., in Proc. IEEE Conference on Computational Imaging Using Synthetic Apertures (CISA), 2024.
- High altitude computational lidar emulation and machine learning reconstruction for Earth sciences. Arce, G. R., Ramirez-Jaime, A., Porras-Diaz, N., in Proc. Big Data VI: Learning, Analytics, and Applications, 2024.
- Nonlinear model predictive control with constraint satisfactions for a quadcopter. Wang, Y., Ramirez-Jaime, A., Xu, F., Puig, V., in Proc. Journal of Physics: Conference Series, 2017.
- A differential game approach to urban drainage systems control. Ramirez-Jaime, A., Quijano, N., Ocampo-Martinez, C., in Proc. American Control Conference (ACC), 2016.
- Co-simulation for the design of controllers in urban drainage systems. Riaño-Briceño, G., Ramirez-Jaime, A., Barreiro-Gomez, J., Quijano, N., Ocampo-Martinez, C., in Proc. IEEE 2nd Colombian Conference on Automatic Control (CCAC), 2015.

References

- Gonzalo R. Arce — Charles Black Evans Professor, Department of Electrical & Computer Engineering, University of Delaware. arce@udel.edu
- Javier Garcia-Frias — Professor and Associate Chair (Graduate Studies), Department of Electrical & Computer Engineering, University of Delaware. jgf@udel.edu
- Volodymyr (Vladimir) Lukin — Head, Department of Information-Communication Technologies, National Aerospace University “Kharkiv Aviation Institute” (KhAI). v.lukin@khai.edu
- Krzysztof Okarma — Professor, Department of Signal Processing and Multimedia Engineering, West Pomeranian University of Technology in Szczecin (ZUT). okarma@zut.edu.pl
- Mark Stephen — Engineer, NASA Goddard Space Flight Center. mark.a.stephen@nasa.gov
- Guangning Yang — Engineer, NASA Goddard Space Flight Center. Guangning.Yang-1@nasa.gov